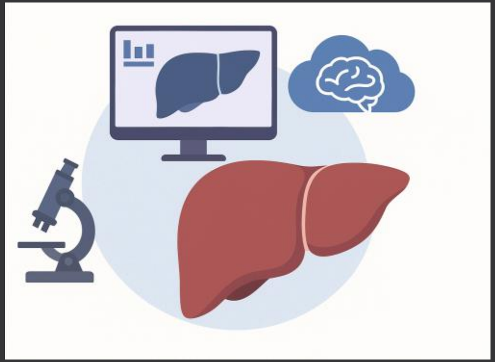
Reducing Diagnostic Errors in Liver Disease Using a Multi-Modal AI Clinical Decision Support System

Group 13 Group Members:

Sanjana Nelli (s4118893)

Neeharika Rimmanapudi (s4118918)

Prathibha Magesh (s3859590)

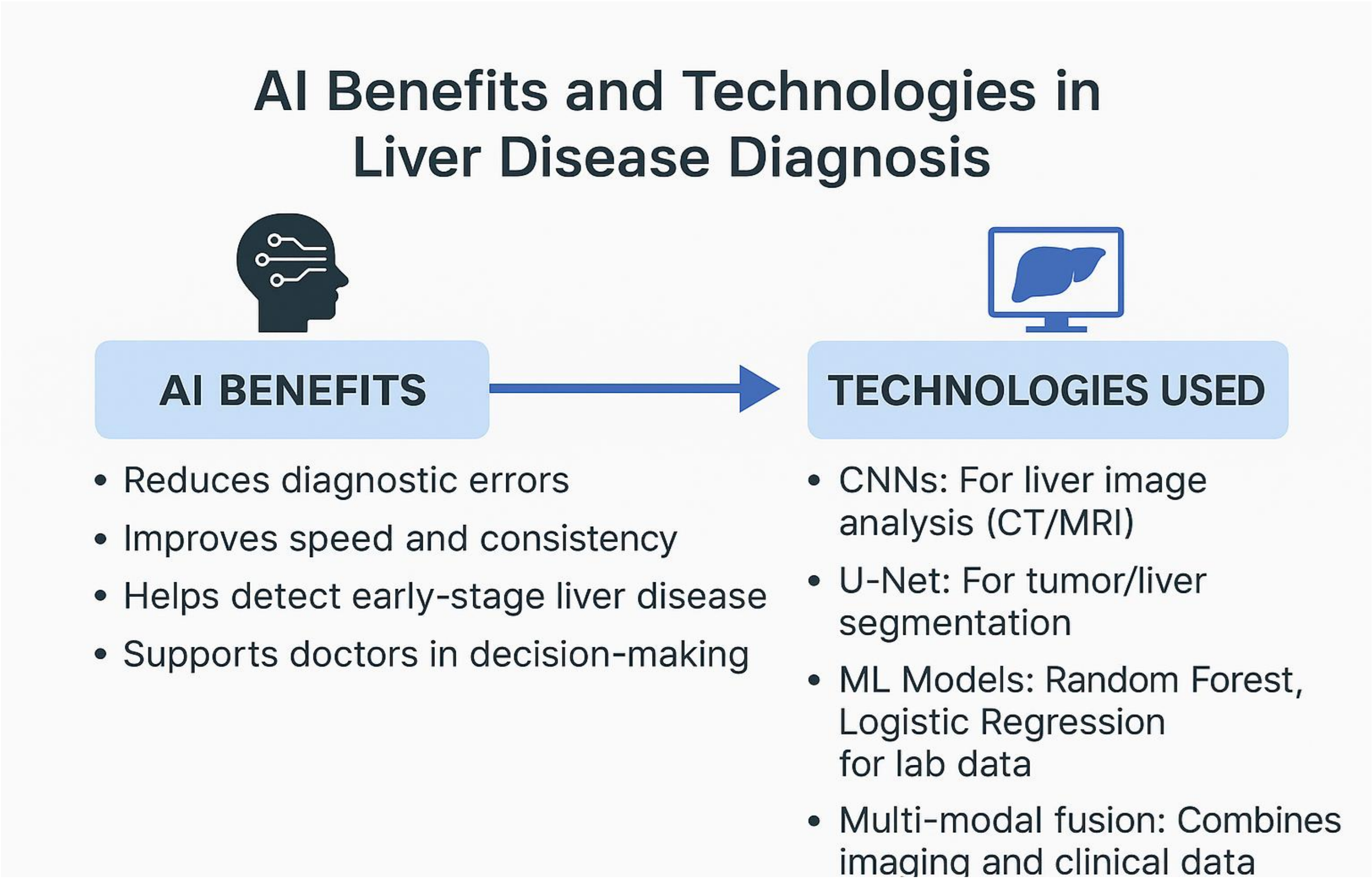


Problem Statement

Manual diagnosis of liver diseases like HCC and cirrhosis is prone to **error due to variability** in expertise, scattered data, and diagnostic overload.

The **World Health Organization** reports that **over 40% of liver tumors are diagnosed at advanced stages**, reducing treatment success and survival rates.

An AI-driven multi-modal Clinical Decision Support System (CDSS) that integrates clinical and imaging data is urgently needed to support early, accurate diagnosis.


AI Benefits and Technologies in Liver Disease Diagnosis
AI BENEFITS
TECHNOLOGIES USED
• Reduces diagnostic errors
• Improves speed and consistency
• Helps detect early-stage liver disease
• Supports doctors in decision-making
• CNNs: For liver image analysis (CT/MRI)
• U-Net: For tumor/liver segmentation
• ML Models: Random Forest, Logistic Regression for lab data
• Multi-modal fusion: Combines imaging and clinical data

Existing Systems

How is Liver Disease currently diagnosed?

Imaging Review

Radiologists manually assess CT, MRI, or ultrasound scans to identify abnormalities.

Clinical Assessment

Doctors evaluate symptoms, medical history, and physical exams to make diagnostic decisions

Laboratory or Genetic Testing

Blood tests or genetic screening are analyzed to confirm disease presence or risk factors

Limitations:

- Diagnosis depends heavily on radiologist and clinician experience, which varies.
- No use of automated segmentation or pattern recognition in imaging.
- Small tumors or slight lab anomalies may go unnoticed.
- Shortage of specialists (e.g., hepatologists, radiologists) in non-urban clinics.
- Manual systems lack feedback loops to improve over time.

LLM-AIDED RESEARCH FOR CDSS IN LIVER DISEASE DIAGNOSIS

ChatGPT Investigation

Using ChatGPT to support CDSS research

- Clarifying key concepts (e.g, CNNs, SHAP AI
- Summarizing research papers on AI in liver diagnosis

AI Models for CDSS

Identifying AI in U-Net, and ensemble models

- CNNs/U-Net for imaging (CT, MRI) segmentation
- Logistic regresiion, Random Forest for lab data

Clinical Applications

Exploring AI in liver disease diagnosis

- Early detection of HCC, NAFLD, cirrhosis
- Supporting radiologists and general practiti-oners

Data Sources

Understanding data needed for AI-based CDSS

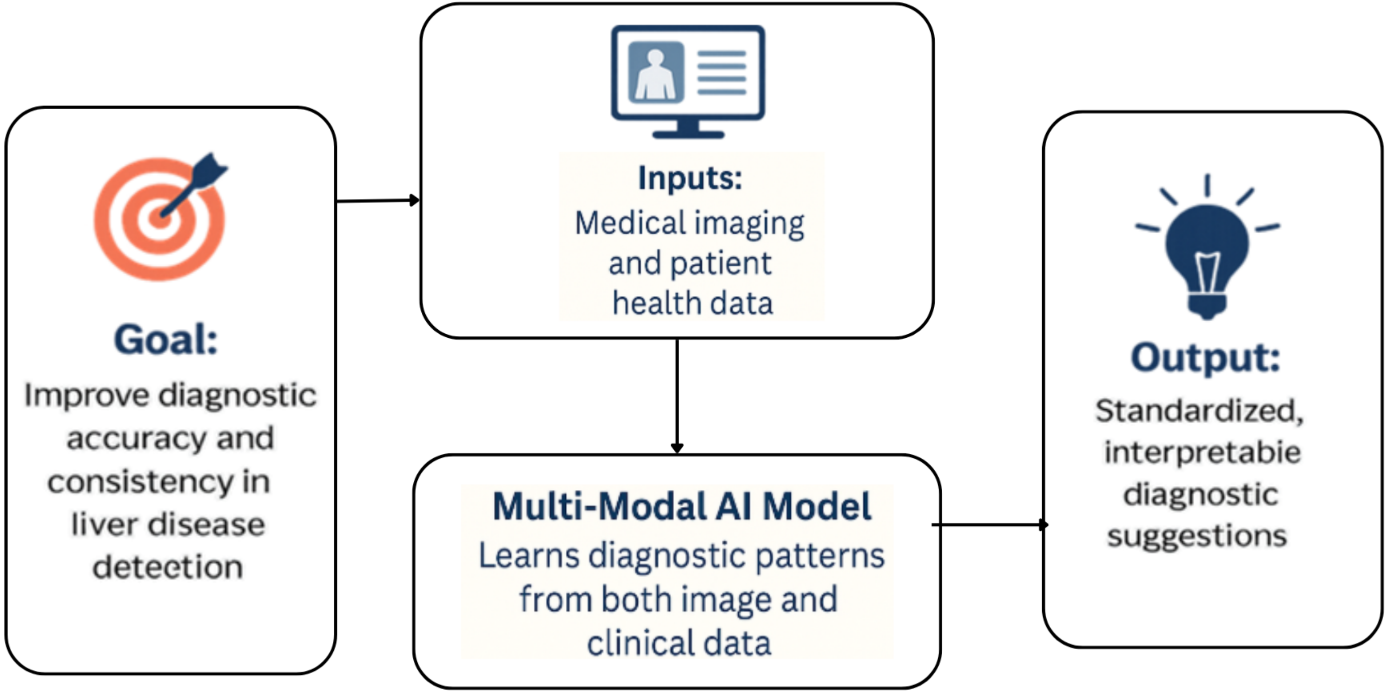
- Imaging data: CT, MRI, ultrasund scans
- Clinical data: ALT, AST, bilirubin, patient history

Challenges

Understanding AI-CDSS implementation challenges

- Data privacy, integration with hospital systems
- Model explainaibility across patientiert-populations

Preliminary Solution Concept



Conclusion

Current liver disease diagnostic systems are prone to errors due to manual interpretation and inconsistent integration of medical data. A **Multi-Modal AI Clinical Decision Support System (CDSS)** offers a more reliable, consistent, and automated approach to combining imaging and clinical data, significantly reducing diagnostic errors and improving early detection of liver diseases like HCC, cirrhosis, and NAFLD.

Next Steps:

1.Complete Detailed Research and Literature Review

2.Refine the Proposed CDSS System Concept

3. Data Collection and Integration

4.System Development and Validation

5. Deployment and Monitoring